

Operations Research

Duality

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Course outline

- **Linear Optimization**

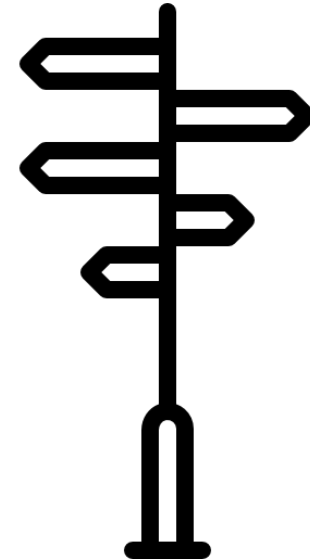
- Modeling by linear programming and graphical solution
- Solving linear programs, convexity
- Simplex algorithm, 2-phase simplex
- Simplex algorithm in matrix notation
- Sensitivity analysis, reoptimization
- **Duality theory, zero-sum games**

- **Integer Linear Optimization**

- Modeling integer programming problems (IPs)
- Solving IPs: branch & bound, cutting planes
- Advanced solution methods: column generation, approximation algorithms
- Fast solvable integer problems: total unimodularity, matroids
- Network flow problems
- Traveling salesperson problem

- **Nonlinear Optimization**

- Optimization of multidimensional functions
- Convex optimization



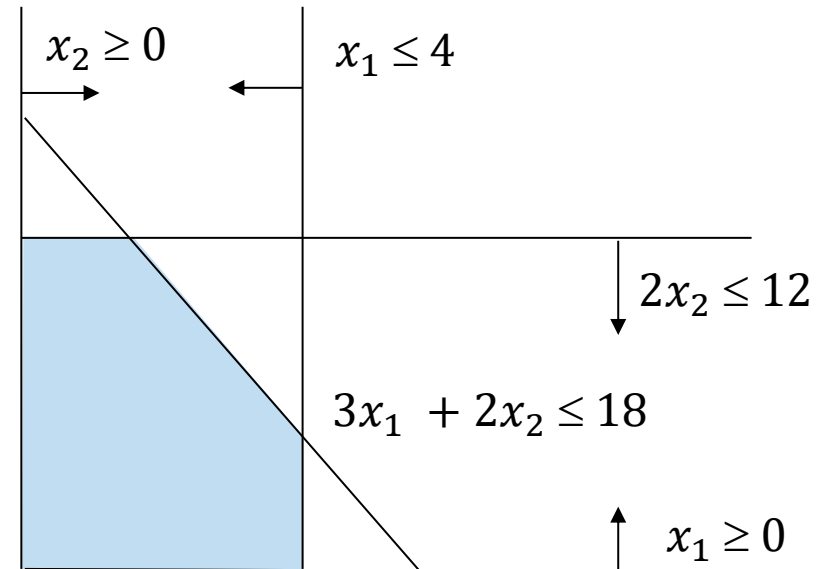
Today you will learn...

- the importance of the **dual problem** for linear programming.
- how to **formulate** dual problems.
- how to derive **optimal strategies in zero-sum games** from shadow prices.



Example from last time

$$\begin{array}{llllll}
 \max & 5x_1 & + & 3x_2 & & \\
 \text{s.t.} & x_1 & & & + & x_3 & = & 4 \\
 & & 2x_2 & & + & x_4 & = & 12 \\
 & 3x_1 & + & 2x_2 & & & + & x_5 & = & 18 \\
 & & & & x_1, x_2, x_3, x_4, x_5 & \geq & 0
 \end{array}$$



The solution of the LP is $(4, 3, 0, 6, 0)$ with basic variables $\{x_1, x_2, x_4\}$ and objective function value **29**.

Recap: shadow prices

$$\begin{array}{llllll} \max & 5x_1 & + & 3x_2 & & \\ \text{s.t.} & x_1 & & & + & x_3 & = & 4 \\ & & 2x_2 & & + & x_4 & = & 12 \\ & 3x_1 & + & 2x_2 & & & + & x_5 & = & 18 \\ & & & & & x_1, x_2, x_3, x_4, x_5 & \geq & 0 \end{array}$$

x_3 is the slack variable corresponding to the capacity constraint of machine 1.

If the capacity is increased from 4 to 5, we obtain the point $(5, 3/2)$ as the optimal solution, with an objective function value of 29.5.

Shadow prices (recap)

The entries under the slack variables contained in the result row of each simplex tableau are called opportunity costs (shadow prices).

Basic variable	Coefficient						Solution
	z	x_1	x_2	x_3	x_4	x_5	
z	1			$1/2$		$3/2$	29.5
x_1		1		1		0	5
x_2			1	$-3/2$		$1/2$	1.5
x_4				3	1	-1	9

$$c'_N = -(c_N - (B^{-1}N)^T c_B)^T = -c_N^T + c_B^T B^{-1}N \text{ and } c_B^T B^{-1} = \left(\frac{1}{2}, 0, \frac{3}{2}\right)$$

$$c'_N = -(0 \ 0) + \left(\frac{1}{2}, 0, \frac{3}{2}\right) \begin{pmatrix} 1 & 0 \\ 0 & 0 \\ 0 & 1 \end{pmatrix} = \left(\frac{1}{2}, \frac{3}{2}\right)$$

Duality

$$\begin{array}{llll} \max & 5x_1 & + & 3x_2 \\ \text{s.t.} & x_1 & & \leq 4 \quad (\text{machine 1}) \\ & & 2x_2 & \leq 12 \quad (\text{machine 2}) \\ & 3x_1 & + & 2x_2 \leq 18 \quad (\text{machine 3}) \\ & x_1, x_2 & \geq & 0 \end{array}$$

Suppose we do not produce ourselves, but can sell the available machines (hours) to another company.

Price per hour on machine i : y_i

How much should the buyer pay for it?

Primal and dual problems

The buyer's problem is also called the **dual problem** of the **primal problem**.

The optimal solution of the dual problem gives the minimum prices that a buyer would have to pay for the resources of our company, such that it makes sense for us to give up on the resources.

General form of the dual problem

For a primal problem:

$$\begin{array}{ll}
 \max & 5x_1 + 3x_2 \\
 \text{s.t.} & x_1 \leq 4 \\
 & 2x_2 \leq 12 \\
 & 3x_1 + 2x_2 \leq 18 \\
 & x_1, x_2 \geq 0
 \end{array}$$

$$\begin{array}{ll}
 \max & c^T x \\
 \text{s.t.} & Ax \leq b \\
 & x \geq 0
 \end{array}$$

... is the dual problem:

$$\begin{array}{ll}
 \min & 4y_1 + 12y_2 + 18y_3 \\
 \text{s.t.} & y_1 + 3y_3 \geq 5 \\
 & 2y_2 + 2y_3 \geq 3 \\
 & y_1, y_2, y_3 \geq 0
 \end{array}$$

$$\begin{array}{ll}
 \min & b^T y \\
 \text{s.t.} & A^T y \geq c \\
 & y \geq 0
 \end{array}$$

Observations

For each constraint in the primal problem, there is one variable in the dual problem.

→ 3 primal constraints $\Rightarrow$ 3 dual variables

For each variable in the primal problem, there is one constraint in the dual problem.

If the primal problem is a maximization problem, then the dual problem is a minimization problem.

Symmetry property

The dual of the dual problem is the primal problem.

$$\begin{array}{ccc} \begin{array}{l} \max \quad c^T x \\ \text{s.t.} \quad Ax \leq b \\ \quad \quad x \geq 0 \end{array} & \begin{array}{c} \xrightarrow{\text{dual}} \\ \xleftarrow{\text{dual}} \end{array} & \begin{array}{l} \min \quad b^T y \\ \text{s.t.} \quad A^T y \geq c \\ \quad \quad y \geq 0 \end{array} \end{array}$$

Weak duality

Theorem:

Let x' be a feasible solution of the primal problem ($Ax' \leq b, x' \geq 0$) and y' be a feasible solution of the dual problem ($A^T y' \geq c, y' \geq 0$). Then $c^T x' \leq b^T y'$.

$$\begin{array}{ll} \max & c^T x \\ \text{s. t.} & Ax \leq b \\ & x \geq 0 \end{array}$$

$$\begin{array}{ll} \min & b^T y \\ \text{s. t.} & A^T y \geq c \\ & y \geq 0 \end{array}$$

Proof: $c^T x' \leq (A^T y')^T x' = y'^T (Ax') \leq b^T y'$

The objective value of the minimization problem is greater than or equal to the objective value of the maximization problem.

The optimal objective value of the dual problem is an **upper bound** for the optimal solution of the primal problem.

(If the primal problem is a minimization problem, then it is a **lower bound**.)

Weak duality: example

$$\begin{array}{llllll}
 \min & 4y_1 & + & 12y_2 & + & 18y_3 \\
 \text{s.t.} & y_1 & & & + & 3y_3 \geq 5 \\
 & & & 2y_2 & + & 2y_3 \geq 3 \\
 & & & & & y_1, y_2, y_3 \geq 0
 \end{array}$$

A **feasible dual solution** is:

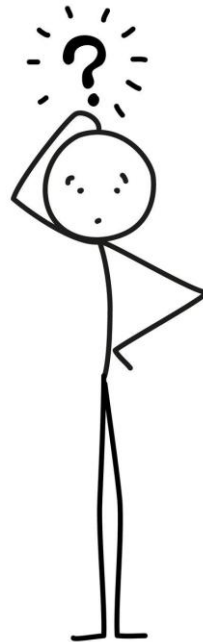
$y_1 = 1, y_2 = 0, y_3 = 2$ with value 40

$$\begin{array}{llll}
 \max & 5x_1 & + & 3x_2 \\
 \text{s.t.} & x_1 & & \leq 4 \\
 & & 2x_2 & \leq 12 \\
 & 3x_1 & + & 2x_2 \leq 18 \\
 & & x_1, x_2 & \geq 0
 \end{array}$$

Optimal primal solution:

$x_1 = 4, x_2 = 3$ with value 29

How are dual variables and shadow prices related?



General scheme

	Primal	Dual
Normal Form	$\begin{aligned} \max \quad & c^T x \\ \text{s.t.} \quad & Ax \leq b \\ & x \geq 0 \end{aligned}$	$\begin{aligned} \min \quad & b^T y \\ \text{s.t.} \quad & A^T y \geq c \\ & y \geq 0 \end{aligned}$
Standard Form	$\begin{aligned} \max \quad & c^T x \\ \text{s.t.} \quad & Ax = b \\ & x \geq 0 \end{aligned}$	$\begin{aligned} \min \quad & b^T y \\ \text{s.t.} \quad & A^T y \geq c \\ & y \text{ unbounded} \end{aligned}$

Complementary slackness

Suppose y^* is the optimal solution of the dual LP and x^* is the optimal solution of the primal LP.

We know from sensitivity analysis:

- If $x_1^* < 4$, then $y_1^* = 0$ (constraint is not binding, shadow prices are 0)
- If $y_1^* > 0$, then $x_1^* = 4$
 $\rightarrow y_1^*(x_1^* - 4) = 0$

Due to the symmetry between primal and dual, it holds: $\rightarrow x_1^*(y_1^* + 3y_3^* - 5) = 0$

$$\begin{array}{ll}
 \max & 5x_1 + 3x_2 \\
 \text{s.t.} & x_1 \leq 4 \\
 & \quad + 2x_2 \leq 12 \\
 & 3x_1 + 2x_2 \leq 18 \\
 & x_1, x_2 \geq 0
 \end{array}$$

$$\begin{array}{ll}
 \min & 4y_1 + 12y_2 + 18y_3 \\
 \text{s.t.} & y_1 + 3y_3 \geq 5 \\
 & 2y_2 + 2y_3 \geq 3 \\
 & y_1, y_2, y_3 \geq 0
 \end{array}$$

Complementary slackness theorem

Given a primal-dual pair:

$$\begin{array}{ll} \max & c^T x \\ \text{s.t.} & Ax \leq b \\ & x \geq 0 \end{array} \quad \begin{array}{ll} \min & b^T y \\ \text{s.t.} & A^T y \geq c \\ & y \geq 0 \end{array}$$

Theorem:

Let x^* and y^* be feasible solutions to the primal and the dual problem, respectively. The vectors x^* and y^* are optimal solutions for the two respective problems if and only if:

$$(A^T y^* - c)^T x^* = 0 \text{ and } (Ax^* - b)^T y^* = 0$$

Dimensions: $A = m \times n$, $y = m \times 1$, $x = n \times 1$

Strong duality

Theorem:

If x^* is an optimal solution of the primal problem and y^* is an optimal solution of the dual problem, then $c^T x^* = b^T y^*$.

Proof: Since x^* and y^* satisfy the conditions of the complementary slackness:

$$(A^T y^* - c)^T x^* = 0 \equiv y^{*T} A x^* = c^T x^*$$

$$(A x^* - b)^T y^* = 0 \equiv (A x^*)^T y^* = b^T y^*$$

$$\text{So } b^T y^* = c^T x^*.$$

Theorem: (Version via feasibility)

Let P and D be a primal-dual pair. If P and D are both feasible, then there exist an optimal solution x^* of P an optimal solution y^* of D and the objective values corresponding to x^* and y^* are equal.

Using the results

If we know an optimal primal solution, we can find an optimal dual solution.

For our example: $x^* = (x_1^* \ x_2^*) = (4, 3)$

$$x_1^* > 0 \Rightarrow y_1^* + 3y_3^* = 5$$

$$x_2^* > 0 \Rightarrow 2y_2^* + 2y_3^* = 3$$

$$2x_2^* < 12 \Rightarrow y_2^* = 0$$

$$y_1^* = 0.5, y_2^* = 0, y_3^* = 1.5$$

This corresponds to the shadow prices of the resources $c_B^T B^{-1}$!

$$\begin{array}{ll} \max & 5x_1 + 3x_2 \\ \text{s. t.} & x_1 \leq 4 \\ & +2x_2 \leq 12 \\ & 3x_1 + 2x_2 \leq 18 \\ & x_1, x_2 \geq 0 \end{array}$$

$$\begin{array}{ll} \min & 4y_1 + 12y_2 + 18y_3 \\ \text{s. t.} & y_1 + 3y_3 \geq 5 \\ & 2y_2 + 2y_3 \geq 3 \\ & y_1, y_2, y_3 \geq 0 \end{array}$$

Strong duality: example

If x^* is an optimal solution of the primal problem and y^* is an optimal solution of the dual problem, then $c^T x^* = b^T y^*$.

$$\begin{aligned} \min \quad & 4y_1 + 12y_2 + 18y_3 \\ \text{s.t.} \quad & y_1 + 3y_3 \geq 5 \\ & 2y_2 + 2y_3 \geq 3 \\ & y_1, y_2, y_3 \geq 0 \end{aligned}$$

The optimal dual solution is:

$$y_1 = 0.5, \quad y_2 = 0, \quad y_3 = 1.5 \text{ with value } 29$$

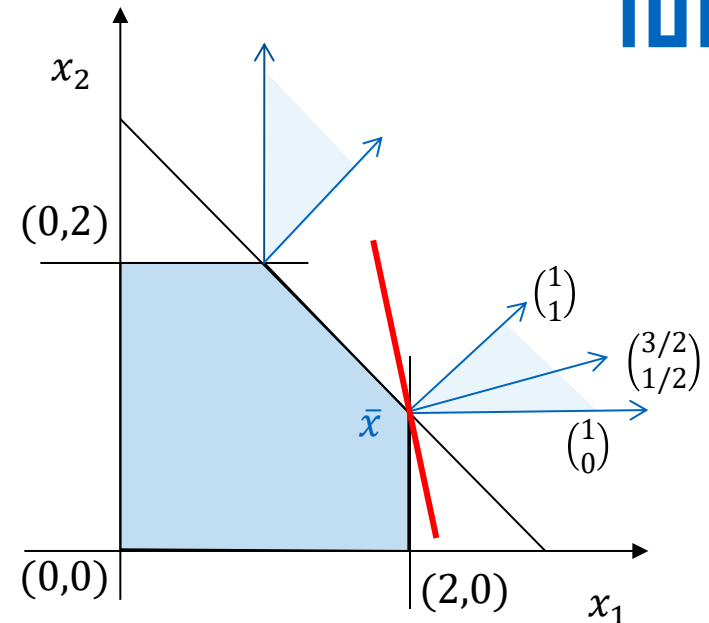
y_i^* are called **Lagrange multipliers, dual variables** or **shadow prices**.

Geometrical interpretation

$$\max \frac{3}{2}x_1 + \frac{1}{2}x_2$$

s. t.

$$\begin{pmatrix} 1 & 0 \\ 1 & 1 \\ 0 & 1 \end{pmatrix} x \leq \begin{pmatrix} 2 \\ 3 \\ 2 \end{pmatrix}, x \text{ unrestricted}$$



A feasible solution is $\bar{x} = (2,1)^T$. Binding for $\bar{x}$ are the 1st and the 2nd constraints.

Therefore, the cone of binding constraints for $\bar{x}$ is $\left\{ \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 1 \\ 1 \end{pmatrix} \right\}$, which you get from A^T .

Theorem:

Let $\bar{x}$ be a feasible solution of the problem $\max\{c^T x : Ax \leq b\}$. Then $\bar{x}$ is optimal if and only if c lies in the cone of binding constraints for $\bar{x}$.

$$\begin{pmatrix} 3/2 \\ 1/2 \end{pmatrix} = \bar{y}_1 \begin{pmatrix} 1 \\ 0 \end{pmatrix} + \bar{y}_2 \begin{pmatrix} 1 \\ 1 \end{pmatrix} \text{ for } \bar{y}_1 = 1 \geq 0 \text{ and } \bar{y}_2 = \frac{1}{2} \geq 0$$

show that $\begin{pmatrix} 3/2 \\ 1/2 \end{pmatrix}$ lies in the cone $\Rightarrow$ Feasibility in the dual:

$$\begin{array}{ll} \min & (2, 3, 2)y \\ \text{s. t.} & \begin{pmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \end{pmatrix} y = \begin{pmatrix} 3/2 \\ 1/2 \end{pmatrix} \\ & y \geq 0 \end{array}$$

Infeasibility / Unboundedness

(Primal)

$$\begin{array}{ll}
 \min & x_1 \\
 \text{s.t.} & x_1 + x_2 \geq 3 \\
 & -x_1 \geq -1 \\
 & -x_2 \geq -1 \\
 & x_1, x_2 \geq 0
 \end{array}$$

No feasible solution

(Dual)

$$\begin{array}{ll}
 \max & 3y_1 - y_2 - y_3 \\
 \text{s.t.} & y_1 - y_2 \leq 1 \\
 & y_1 - y_3 \leq 0 \\
 & y_1, y_2, y_3 \geq 0
 \end{array}$$

Unbounded

Infeasibility / Unboundedness

Theorem:

If the primal LP is unbounded, then the dual problem is infeasible.

Theorem:

If the dual LP is unbounded, then the primal problem is infeasible.



Julius
Farkas
1847-1930

Farkas' lemma

Lemma (Farkas, 1902): Given a real $m \times n$ matrix A and a real m -dimensional vector b . Exactly one of the statements below holds (component-wise):

1. $Ax = b, x \geq 0$
2. $A^T y \leq 0, b^T y > 0$ (unbounded)

Proof: The proof is based on strong duality. Consider the following primal and dual LP:

$$\begin{aligned} \min \quad & 0x \\ \text{s.t.} \quad & Ax = b \\ & x \geq 0 \end{aligned}$$

$$\begin{aligned} \max \quad & b^T y \\ \text{s.t.} \quad & A^T y \leq 0 \\ & y \text{ unconstrained} \end{aligned}$$

If (1.) holds (i.e., the primal is feasible), then the objective function value of the primary LP is 0. The primal optimum 0 is an upper bound for the dual objective function value.

It follows that $b^T y > 0$ cannot hold.

Conversely, if (1.) does not hold, then the primary LP is infeasible.

We know from strong duality that the dual LP is then unbounded.

If the dual were bounded and feasible, it would contradict the infeasibility of the primal.

Farkas' lemma by example

Check $Ax=b$ with $x \geq 0$

$$x_1 + 2x_2 = 1$$

$$-x_1 + 2x_2 = 1$$

We have $x_1 = 0$ and $x_2 = 0.5$, which are non-negative.

Now $y^T A \leq 0$:

$$y_1 - y_2 \leq 0$$

$$2y_1 + 2y_2 \leq 0$$

$y^T b > 0$:

$$y_1 + y_2 > 0$$

Check:

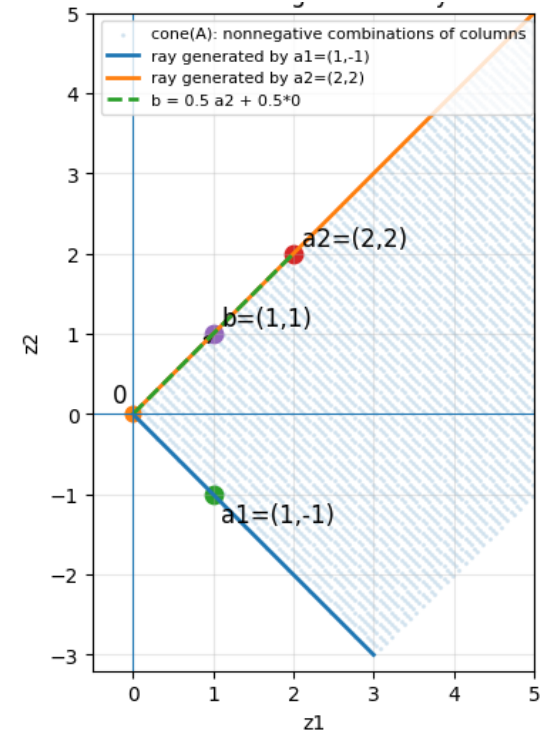
$y_1 - y_2 \leq 0$ implies $y_1 \leq y_2$.

$2y_1 + 2y_2 \leq 0$ simplifies to $y_1 + y_2 \leq 0$.

But $y_1 + y_2 > 0$ is needed for the second part.

Contradiction!

$b = (1,1)$ is inside the cone $(1,-1)$ and $(2,2)$.



Farkas' lemma by example

Check $Ax=b$ with $x \geq 0$

$$\begin{aligned} x_1 + 2x_2 &= 0 \\ -x_1 + 2x_2 &= 2 \end{aligned}$$

which is infeasible, because $x_1 = -1$.

Now $y^T A \leq 0$:

$$\begin{aligned} y_1 - y_2 &\leq 0 \\ 2y_1 + 2y_2 &\leq 0 \end{aligned}$$

$y^T b > 0$:

$$2y_2 > 0$$

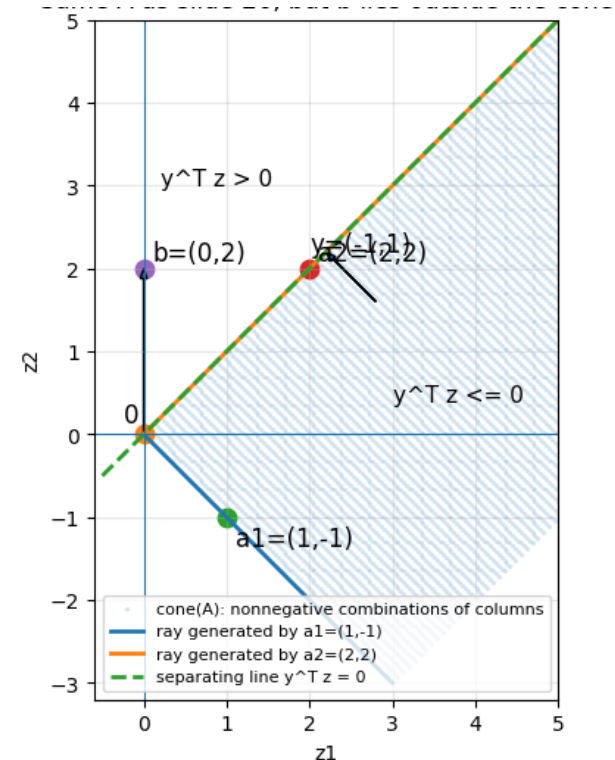
Check:

$y_1 - y_2 \leq 0$ implies $y_1 \leq y_2$.

$2y_1 + 2y_2 \leq 0$ simplifies to $y_1 + y_2 \leq 0$.

$\Rightarrow y_1 \leq y_2, y_1 \leq -y_2, \max 2y_2 \Rightarrow$ unbounded

$b = (1,1)$ is outside the cone $(1,-1)$ and $(2,2)$.



Summary duality

For each LP, there is always a dual LP; one problem is a maximization, the other problem is a minimization.

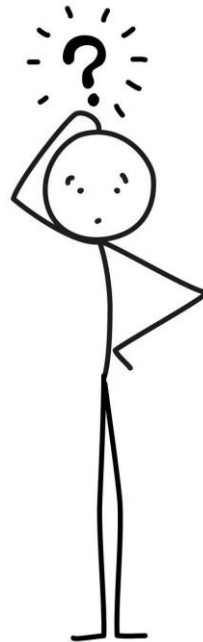
Weak duality:

- In a primal-dual pair, the value of each solution of the maximization problem is less than or equal to the value of each solution of the minimization problem.

Strong duality:

- Suppose B is an optimal basis of the primal LP. Then $y^T = c_B^T B^{-1}$ is an optimal solution of the dual problem, i.e. the primal and the dual objective values are equal.
- If B is a feasible basis of the primal LP and $y^T = c_B^T B^{-1}$ is feasible for the dual problem, then B (and y) are optimal.

What is complementary slackness?



Further applications of duality

Optimal strategies in games → Zero-sum games in matrix form

Pricing in auctions → Assignment markets

Zur Theorie der Gesellschaftsspiele¹⁾.

Von

J. v. Neumann in Berlin.

Einleitung.

1. Die Frage, deren Beantwortung die vorliegende Arbeit anstrebt, ist die folgende:

n Spieler, $S_1, S_2, \dots, S_n$, spielen ein gegebenes Gesellschaftsspiel $\mathcal{G}$. Wie muß einer dieser Spieler, S_m , spielen, um dabei ein möglichst günstiges Resultat zu erzielen?

Die Fragestellung ist allgemein bekannt, und es gibt wohl kaum eine Frage des täglichen Lebens, in die dieses Problem nicht hineinspielt; trotzdem ist der Sinn dieser Frage kein eindeutig klarer. Denn sobald $n > 1$ ist (d. h. ein eigentliches Spiel vorliegt), hängt das Schicksal eines jeden Spielers außer von seinen eigenen Handlungen auch noch von denen seiner Mitspieler ab; und deren Benehmen ist von genau denselben egoistischen Motiven beherrscht, die wir beim ersten Spieler bestimmen möchten. Man fühlt, daß ein gewisser Zirkel im Wesen der Sache liegt.

Wir müssen also versuchen, zu einer klaren Fragestellung zu kommen. Was ist zunächst ein Gesellschaftsspiel? Es fallen unter diesen Begriff sehr viele, recht verschiedenartige Dinge: von der Roulette bis zum Schach, vom Bakkarat bis zum Bridge liegen ganz verschiedene Varianten des Sammelbegriffes „Gesellschaftsspiel“ vor. Und letzten Endes kann auch irgendein Ereignis, mit gegebenen äußeren Bedingungen und gegebenen Handelnden (den absolut freien Willen der letzteren vorausgesetzt), als Gesellschaftsspiel angesehen werden, wenn man seine Rückwirkungen auf die in ihm handelnden Personen betrachtet²⁾. Was ist nun das gemeinsame Merkmal aller dieser Dinge?

¹⁾ Der Inhalt dieser Arbeit ist (mit einigen Kürzungen) am 7. XII. 1926 der Göttinger Math. Ges. vorgetragen worden.

²⁾ Es ist das Hauptproblem der klassischen Nationalökonomie: was wird, unter gegebenen äußeren Umständen, der absolut egoistische „homo oeconomicus“ tun?

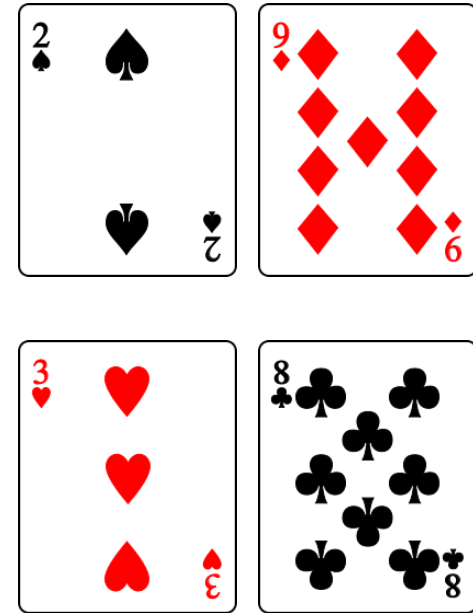
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I want to play a game

A fellow student suggests the following game to you:

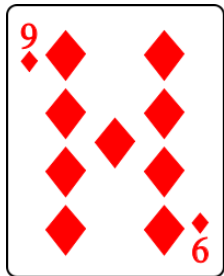
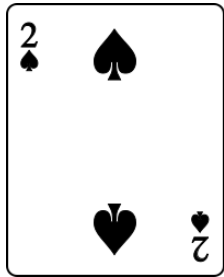
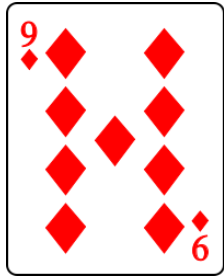
- You receive two playing cards, the 2 of spades and the 9 of diamonds.
- Your fellow student receives the 3 of hearts and the 8 of clubs



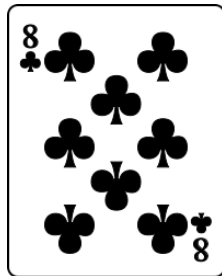
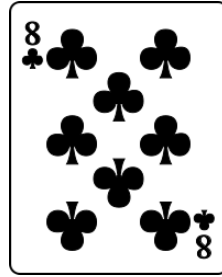
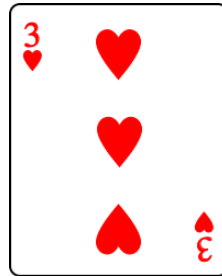
- You both choose one card secretly and then lay it face up at the same time.
- If both cards are of **the same color** (black or red), then your fellow student pays you as many Euros as the value of your card.
- If the cards are of **different colors**, (one black, one red), you pay your fellow student as many Euros as the value of his card.

Example

You



Fellow student



Solution



Which type of game do we have?

In game theoretic terminology, you are playing a static zero-sum game with complete information.

Static:

Players choose their strategy (action) without knowing what the other players are choosing.

Complete information:

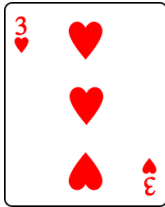
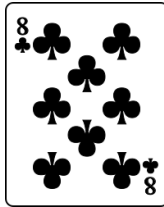
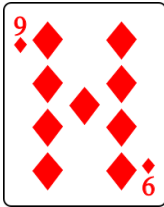
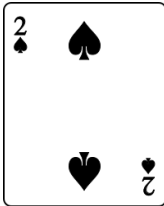
Each player knows exactly what are the payoffs in the end once all strategies are chosen.

Zero-sum (for two players):

One player's win is equal to the other player's loss.

Matrix games

Let's get back to our game from before:

		Possible strategies of the fellow student	
			
Your possible strategies		+9 €	-8 €
		-3 €	+2 €

Your profit

Zero-sum games in matrix form (for 2 persons)

More abstract:

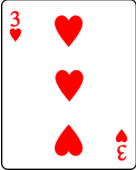
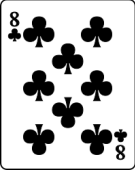
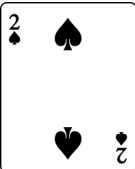
		Actions player B (column player)			
		s_{B1}	s_{B2}	$\dots$	s_{Bn}
Actions player A (row player)	s_{A1}	$u_A(s_{A1}, s_{B1})$	$\dots$	$\dots$	$u_A(s_{A1}, s_{Bn})$
	s_{A2}	$\vdots$	$\ddots$	$\ddots$	$\vdots$
	$\vdots$	$\vdots$	$\ddots$	$\ddots$	$\vdots$
	s_{Am}	$u_A(s_{Am}, s_{B1})$	$\dots$	$\dots$	$u_A(s_{Am}, s_{Bn})$

$u_A(s_{Am}, s_{B1})$: Utility of player A,
when A plays strategy s_{Am} and
B plays strategy s_{B1} .

Optimal strategy as an LP

Possible strategies of the fellow student

Your possible strategies

		
	+9	-8
	-3	+2

$$\max g$$

$$\text{s.t.} \quad 9x_1 - 3x_2 \geq g \quad (\text{Opponent plays heart 3})$$

$$-8x_1 + 2x_2 \geq g \quad (\text{Opponent plays cross 8})$$

$$x_1 + x_2 = 1$$

$$x_1 \geq 0, x_2 \geq 0, g \in \mathbb{R}$$

You maximize the minimum profit.

Transform in normal form before dualizing:

$$\max g^+ - g^-$$

$$\text{s.t.} \quad g^+ - g^- - 9x_1 + 3x_2 \leq 0$$

$$g^+ - g^- + 8x_1 - 2x_2 \leq 0$$

$$x_1 + x_2 \leq 1$$

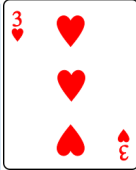
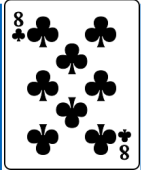
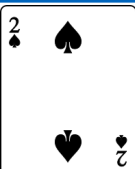
$$-x_1 - x_2 \leq -1$$

$$x_1, x_2, g^+, g^- \geq 0$$

Optimal strategy as LP

Possible strategies of the fellow student

Your possible strategies

		
	+9	-8
	-3	+2

$$\begin{aligned}
 \max \quad & g^+ - g^- \\
 \text{s.t.} \quad & g^+ - g^- - 9x_1 + 3x_2 \leq 0 \\
 & g^+ - g^- + 8x_1 - 2x_2 \leq 0 \\
 & x_1 + x_2 \leq 1 \\
 & -x_1 - x_2 \leq -1 \\
 & x_1, x_2, g^+, g^- \geq 0
 \end{aligned}$$

y_1
 y_2
 π^+
 π^-

Dual Problem:

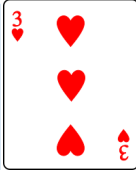
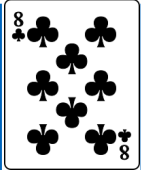
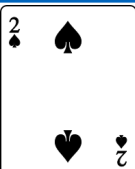
$$\begin{aligned}
 \min \quad & \pi^+ - \pi^- \\
 \text{s.t.} \quad & \pi^+ - \pi^- - 9y_1 + 8y_2 \geq 0 \\
 & \pi^+ - \pi^- + 3y_1 - 2y_2 \geq 0 \\
 & y_1 + y_2 \geq 1 \\
 & -y_1 - y_2 \geq -1 \\
 & y_1, y_2, \pi^+, \pi^- \geq 0
 \end{aligned}$$

x_1
 x_2
 g^+
 g^-

Optimal strategy as LP

Possible strategies of the fellow student

Your possible strategies

		
	+9	-8
	-3	+2

$$\begin{aligned}
 \min \quad & \pi^+ - \pi^- \\
 \text{s.t.} \quad & \pi^+ - \pi^- - 9y_1 + 8y_2 \geq 0 \\
 & \pi^+ - \pi^- + 3y_1 - 2y_2 \geq 0 \\
 & y_1 + y_2 \geq 1 \\
 & -y_1 - y_2 \geq -1 \\
 & y_1, y_2, \pi^+, \pi^- \geq 0
 \end{aligned}$$

Or:

$$\begin{aligned}
 \min \quad & \pi \\
 \text{s.t.} \quad & 9y_1 - 8y_2 \leq \pi \\
 & -3y_1 + 2y_2 \leq \pi \\
 & y_1 + y_2 = 1 \\
 & y_1, y_2 \geq 0, \pi \in \mathbb{R}
 \end{aligned}$$

This is exactly the optimization problem of your opponent who wants to minimize your profit!

You can determine your optimal strategy (and that of your opponent) by solving the primal and dual optimization problems!

Strong duality

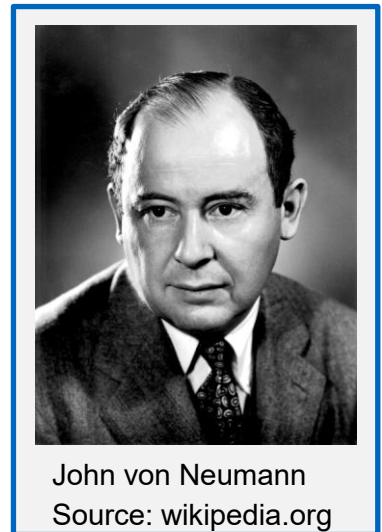
$$\begin{array}{ll}
 \max & g \\
 \text{s.t.} & 9x_1 - 3x_2 \geq g \\
 & -8x_1 + 2x_2 \geq g \\
 & x_1 + x_2 = 1 \\
 & x_1 \geq 0, x_2 \geq 0, g \in \mathbb{R} \\
 & x^* = (0.227, 0.772)
 \end{array}$$

$$\begin{array}{ll}
 \min & \pi \\
 \text{s.t.} & 9y_1 - 8y_2 \leq \pi \\
 & -3y_1 + 2y_2 \leq \pi \\
 & y_1 + y_2 = 1 \\
 & y_1, y_2 \geq 0, \pi \in \mathbb{R} \\
 & y^* = (0.454, 0.545)
 \end{array}$$

The objective function values of both LPs are equal.

The existence of a Nash equilibrium follows from strong LP duality.

The minimax theorem was proposed by von Neumann (1928) and is considered the birth of game theory.



Formulations for both players

Row Player

$$\begin{aligned}
 &\max \quad g \\
 &\text{s.t.} \quad 9x_1 - 3x_2 \geq g \\
 &\quad \quad -8x_1 + 2x_2 \geq g \\
 &\quad \quad x_1 + x_2 = 1 \\
 &\quad \quad x_1 \geq 0, x_2 \geq 0, g \in \mathbb{R} \\
 &\quad \quad x^* = (0.227, 0.772)
 \end{aligned}$$

$$\begin{aligned}
 &\min \quad \pi \\
 &\text{s.t.} \quad 9y_1 - 8y_2 \leq \pi \\
 &\quad \quad -3y_1 + 2y_2 \leq \pi \\
 &\quad \quad y_1 + y_2 = 1 \\
 &\quad \quad y_1, y_2 \geq 0, \pi \in \mathbb{R} \\
 &\quad \quad y^* = (0.454, 0.545)
 \end{aligned}$$

Column Player

$$\begin{aligned}
 &\max \quad g \\
 &\text{s.t.} \quad -9x_1 + 8x_2 \geq g \\
 &\quad \quad 3x_1 - 2x_2 \geq g \\
 &\quad \quad x_1 + x_2 = 1 \\
 &\quad \quad x_1 \geq 0, x_2 \geq 0, g \in \mathbb{R} \\
 &\quad \quad x^* = (0.454, 0.545)
 \end{aligned}$$

$$\begin{aligned}
 &\min \quad \pi \\
 &\text{s.t.} \quad -9y_1 + 3y_2 \leq \pi \\
 &\quad \quad 8y_1 - 2y_2 \leq \pi \\
 &\quad \quad y_1 + y_2 = 1 \\
 &\quad \quad y_1, y_2 \geq 0, \pi \in \mathbb{R} \\
 &\quad \quad y^* = (0.227, 0.772)
 \end{aligned}$$

```
from ortools.linear_solver import pywraplp

def solve_nash_equilibrium():
    solver = pywraplp.Solver.CreateSolver("GLOP")

    # Variables
    p1 = solver.NumVar(0, 1, "p1")
    p2 = solver.NumVar(0, 1, "p2")
    g = solver.NumVar(-100, 100, "g")

    # Objective & constraints
    solver.Maximize(g)
    c1 = solver.Add(9*p1 - 3*p2 >= g)
    c2 = solver.Add(-8*p1 + 2*p2 >= g)
    solver.Add(p1 + p2 == 1)

    if solver.Solve() == pywraplp.Solver.OPTIMAL:
        print("Optimal strategy for Row Player:")
        print(f" Action 1 Probability: {p1.solution_value():.4f}")
        print(f" Action 2 Probability: {p2.solution_value():.4f}")
        print(f" Expected Payoff: {g.solution_value():.4f}")
        print("Optimal strategy for Column Player:")
        print(f" Action 1 Probability: {c1.dual_value():.4f}")
        print(f" Action 2 Probability: {c2.dual_value():.4f}")
    else:
        print("No optimal solution found.")

if __name__ == "__main__":
    solve_nash_equilibrium()
```

Optimal objective -2.727272727e-01

Optimal strategy for Row Player:

Action 1 Probability: 0.22727272727272727

Action 2 Probability: 0.7727272727272727

Minimum Expected Payoff: -0.27272727272

Optimal strategy for Column Player:

Action 1 Probability: -0.45454545454545453

Action 2 Probability: -0.5454545454545454

The expected value of the game in equilibrium is not necessarily zero.

Expected payoff

$$\begin{aligned}
 &\max \quad g \\
 &\text{s.t.} \quad 9x_1 - 3x_2 \geq g \\
 &\quad \quad -8x_1 + 2x_2 \geq g \\
 &\quad \quad x_1 + x_2 = 1 \\
 &\quad \quad x_1 \geq 0, x_2 \geq 0, g \in \mathbb{R} \\
 &\quad \quad x^* = (0.227, 0.772)
 \end{aligned}$$

$$\begin{aligned}
 x^T A y &= (x_1 \ x_2) \begin{pmatrix} 9 & -3 \\ -8 & 2 \end{pmatrix} \begin{pmatrix} y_1 \\ y_2 \end{pmatrix} \\
 &= (0.23 \ 0.77) \begin{pmatrix} 9 & -8 \\ -3 & 2 \end{pmatrix} \begin{pmatrix} 0.45 \\ 0.54 \end{pmatrix} = -0.27
 \end{aligned}$$

$$\begin{aligned}
 &\min \quad \pi \\
 &\text{s.t.} \quad 9y_1 - 8y_2 \leq \pi \\
 &\quad \quad -3y_1 + 2y_2 \leq \pi \\
 &\quad \quad y_1 + y_2 = 1 \\
 &\quad \quad y_1, y_2 \geq 0, \pi \in \mathbb{R} \\
 &\quad \quad y^* = (0.454, 0.545)
 \end{aligned}$$

$$\begin{aligned}
 y^T (-A^T) x &= (y_1 \ y_2) \begin{pmatrix} -9 & 8 \\ 3 & -2 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} \\
 &= (0.45 \ 0.54) \begin{pmatrix} -9 & 3 \\ 8 & -2 \end{pmatrix} \begin{pmatrix} 0.23 \\ 0.77 \end{pmatrix} = 0.27
 \end{aligned}$$

Minimax theorem

Max-min (safety option)

The maximum you can guarantee if the other player is out to harm you.

$$\overline{u}_1 = \max_{x_1} \left(\min_{x_2} u_1(x_1, x_2) \right)$$

Min-max (punishment option)

The lowest the opponent can force your payoff despite your best response.

$$\underline{u}_1 = \min_{x_2} \left(\max_{x_1} u_1(x_1, x_2) \right)$$

Theorem:

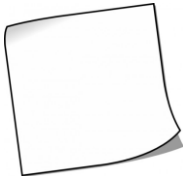
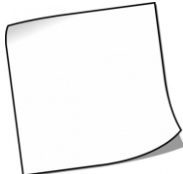
In a finite 2-player zero-sum game, every Nash equilibrium is both a max-min and a min-max of each player.

$$\max_{x_1} \left(\min_{x_2} u_1(x_1, x_2) \right) = \min_{x_2} \left(\max_{x_1} u_1(x_1, x_2) \right)$$

*As far as I can see, there could be no theory of games ... without that theorem ...
I thought there was nothing worth publishing until the Minimax Theorem was proved.*

John von Neumann

A fair game

```

from ortools.linear_solver import pywraplp

def solve_nash_equilibrium():
    solver = pywraplp.Solver.CreateSolver("GLOP")

    # Variables
    pi          = solver.NumVar(-solver.infinity(), solver.infinity(), "pi")
    rock         = solver.NumVar(0, 1, "rock")
    paper        = solver.NumVar(0, 1, "paper")
    scissors     = solver.NumVar(0, 1, "scissors")

    # Objective & constraints as linear expressions
    solver.Maximize(rock + paper + scissors)
    solver.Add(rock + paper + scissors == 1)
    solver.Add(-paper + scissors >= pi)      # payoff_rock
    solver.Add(rock - scissors >= pi)        # payoff_paper
    solver.Add(-rock + paper >= pi)          # payoff_scissors

    if solver.Solve() == pywraplp.Solver.OPTIMAL:
        print(f"  Rock:      {rock.solution_value():.4f}")
        print(f"  Paper:     {paper.solution_value():.4f}")
        print(f"  Scissors: {scissors.solution_value():.4f}")
        print(f"  Payoff:    {pi.solution_value():.4f}")
    else:
        print("No optimal solution found.")

if __name__ == "__main__":
    solve_nash_equilibrium()

```

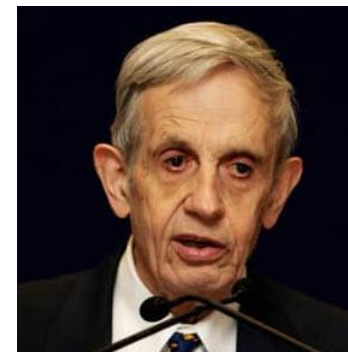
Optimal objective -0.000000000e+00
 Optimal strategy:
 Rock probability: 0.3333333333333333
 Paper probability: 0.3333333333333333
 Scissors probability: 0.3333333333333333

What about general-sum games?

- The Battle of the Sexes
 - Choices: “Lethal Weapon (LW)” vs. “Wondrous Love (WL)”
 - Both prefer to go together
 - Wife prefers LW , husband WL

		<i>Husband</i>	
		LW	WL
<i>Wife</i>	LW	2,1	0,0
	WL	0,0	1,2

- **Nash Equilibrium:** none of the players has an incentive to deviate unilaterally from his strategy.
- Equilibrium problems generalize optimization problems:
 - $u_1(x_1, x_2)$ and $u_2(x_1, x_2)$



John Nash (1928-2015)

History

- von Neumann's original work (1928) used Brouwer's fixed point theorem from 1912 for two-player games. This was before LP duality was developed. von Neumann's theorem (1928) left open the existence of equilibria in general games.
- This was established by John Nash in 1950 using Kakutani's fixed point theorem for correspondences. In 1951 Nash published a proof using Brouwer's fixed point theorem.
- Daskalakis, Goldberg, and Papadimitrou show in 2006 that finding NE is PPAD-hard.

